

ENVX2001 Lab 07 - Regression modelling

ENVX2001 Applied Statistical Methods

Contents

Welcome!	1
Learning outcomes	2
Specific goals	2
Preparation	2
plot()	3
check_model() from performance	3
Downloads	4
Simple linear regression model fitting and interpretation (~ 30 minutes)	4
Analysis	5
Exercise 1	5
1. Exploratory data analysis	5
1.2 State hypotheses	7
1.3 Fit a simple linear regression model	7
1.4 Check model assumptions	10
1.5 Interpret model parameters and fit statistics	11
1.6 Make conclusions	11
Exercise 2: California streamflow	11
Partial F-Tests	12
Question 1	12
Question 2	12
Question 3	12
Question 4	12
Question 5	12
Question 6	13
Review	13
Attribution	13

Welcome!

This week's lab will give you some practice fitting and interpreting Simple Linear Regression (SLR) and Multiple Linear Regression (MLR) models.

This will prepare you for next week's content where we will seek to improve our models, and then in week 9 we will use improved models for prediction.

Learning outcomes

This module links to the following unit learning outcomes:

- LO3. demonstrate proficiency in the use of the statistical programming language R to apply an ANOVA and fit regression models to experimental data
- LO5. interpret the output and understand conceptually how it's derived of a regression, ANOVA and multivariate analysis that have been calculated by R
- LO6. write statistical and modelling results as part of a scientific report

Specific goals

By the end of this lab, students should be able to:

- ☐ Fit a simple linear regression model
- ☐ Assess model assumptions and apply transformations when needed
- ☐ Interpret model parameters and fit statistics
- ☐ Fit a multiple linear regression model (**different structure**)
- ☐ Assess model assumptions, recalling key differences between MLR and SLR (**collinearity**)
- ☐ Determine the most suitable fit statistics for assessing the fit of a MLR model (**r-sq vs adj**)
- ☐ Write a conclusion on your findings



Tip

Please work on this exercise by creating your own R Markdown file.

Preparation

- ☐ Install or update the performance package

CODE

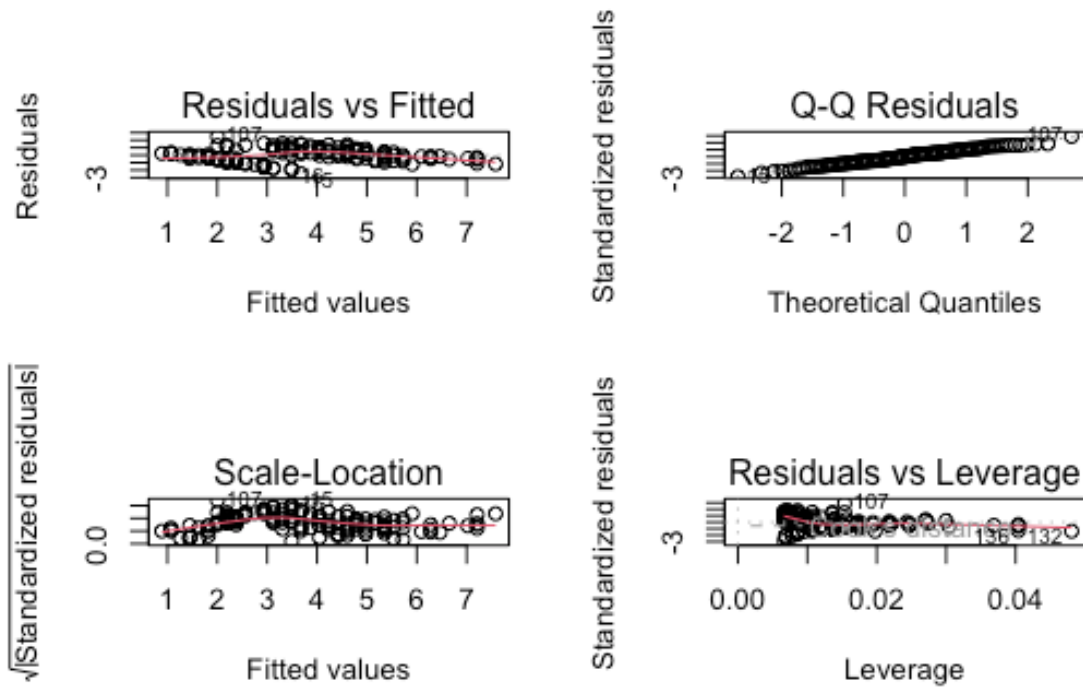
```
#install.packages("performance")
library(performance)
```

This package is really good for checking your models. For this lab, we will focus on the `check_model()` function, which gives us nice pretty diagnostic plots for models:

`plot()`

CODE

```
par(mfrow=c(2,2))  
plot(iris_lm)
```



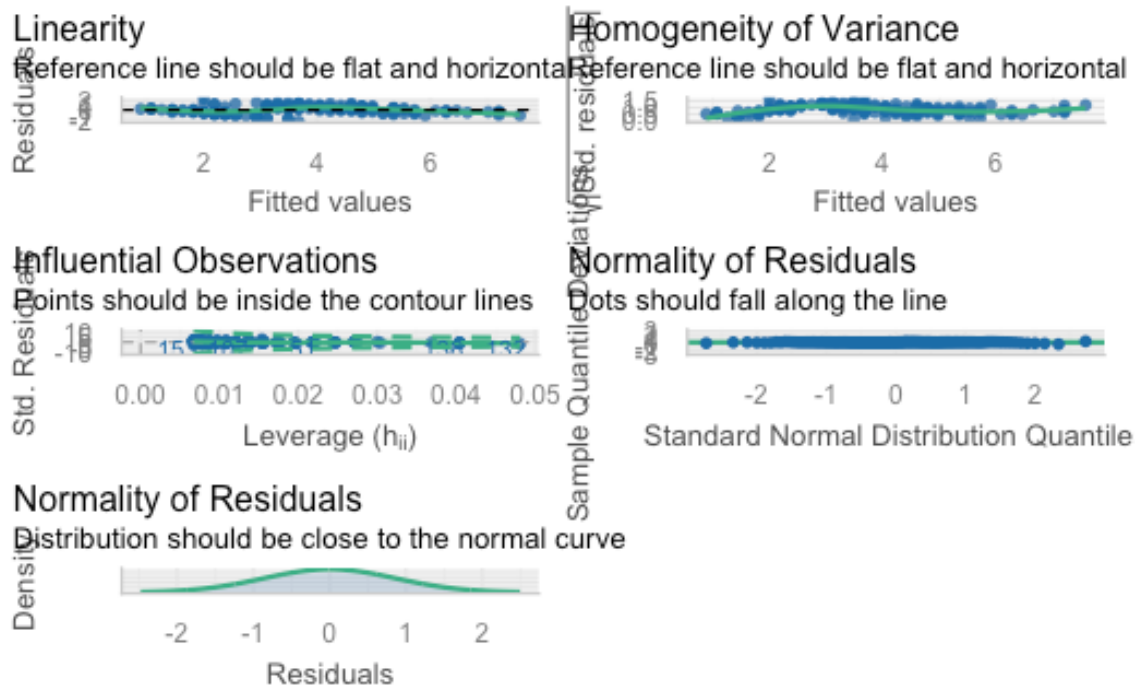
CODE

```
par(mfrow=c(1,1))
```

`check_model()` from `performance`

CODE

```
library(performance)  
check_model(iris_lm)
```



Downloads

File	Used in	Download
Loyn_SLR.xlsx	Exercise 1	Download
california_streamflow.xlsx	Exercise 2	Download

Simple linear regression model fitting and interpretation (~ 30 minutes)

Habitat fragmentation resulting from agriculture and urbanisation has increased the rate of wildlife population decline in Australia. You are a consultant and have been asked to investigate the influence of forest fragment size upon bird abundance. You have been provided with some data containing data on bird abundance and patch area from 56 different forest patches in Southeastern Victoria, Australia, and you want to know if there is a relationship between these two variables.

The data is in the **Loyn_SLR.xlsx** file.

Variables are as follows:

- ABUND: Bird abundance within each forest patch (number of birds)
- AREA: Area of each forest patch (ha)

Original data source: Loyn, R. H. (1987). Effects of patch area and habitat on bird abundances, species numbers, and tree health in fragmented Victorian forests. In *Nature Conservation: the role of remnants of native vegetation* eds. D.S. Saunders, G.W. Arnold, A.A. Burbridge, & A.J.M. Hopkins, pp. 65-77. Surrey Beatty and Sons.

Analysis

Exercise 1

(i) Write out the statistical model for a simple linear regression and define it in terms of influence of forest fragment size upon bird abundance. You can write it out in words and/or in mathematical notation.

(ii) State the hypotheses you will be testing for the relationship between bird abundance and patch area.

(iii) Conduct some exploratory data analysis upon your two variables. Describe the spread and central tendency of each, supported by values (summary statistics) and plots.

```
CODE
# Answer here
```

(iv) Look at the relationship between bird abundance and patch area. Describe the relationship, supported by values (correlation coefficient) and plots.

(v) Fit a simple linear regression model to the data, with bird abundance as the response variable and patch area as the predictor variable.

(vi) Check the assumptions of your model. Do you need to transform any variables?

Hint: you can use the `check_model()` function from the `performance` package to check the assumptions of your model.

(vii) Interpret the model parameters and fit statistics. Recall the key information we draw from the output; what do these statistics tell you about the relationship between bird abundance and patch area?

(viii) Write a concluding statement on your findings. Hint: Consider the initial aim and hypothesis of the study and write it in a way that is easily understood by your client.

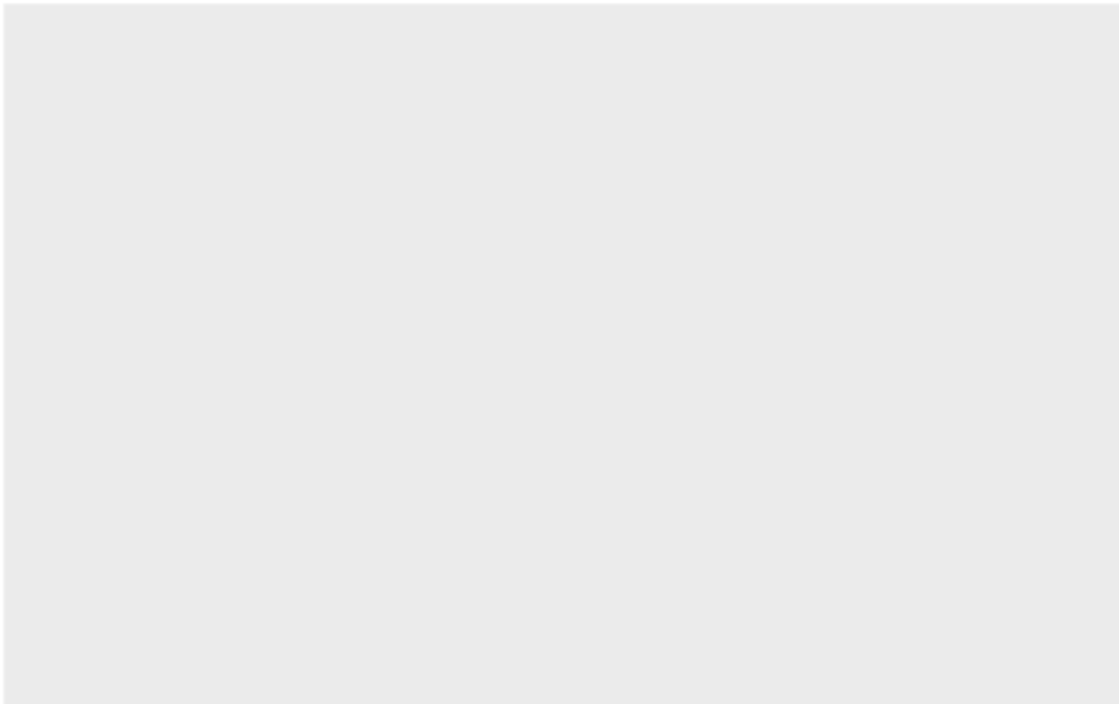
...

1. Exploratory data analysis

We need to understand the data we are working with, so we need to perform exploratory data analysis. To do this, we can look at summary statistics, distributions, and visualise the data.

For linear regression, we also need to look at the relationship between the predictor and response. We first look at a scatter plot to determine if we have linear data, and then we choose a suitable correlation coefficient.

```
CODE
ggplot() +
  geom_point(aes(x = loyn$AREA, y = loyn$ABUND) +
  labs(title = "Bird abundance vs area of forest patch",
        x = "Area of forest patch (ha)",
        y = "Bird abundance (number of birds)"))
```



```
CODE
cor(loyn$AREA, loyn$ABUND) |> round(2) # calculate correlation > round to 2 decimal places
```

```
OUTPUT
[1] 0.26
```

The correlation coefficient is 0.26, which is quite low, indicating a weak or no positive relationship between bird abundance and patch area.

However, we can see from the scatter plot that there is a lot of variation in bird abundance for small patch areas, and that there are some large patches with high bird abundance. This suggests that there may be other factors influencing bird abundance, and that patch area alone may not be a strong predictor. We will explore this further in next week's lecture.

For now, it is possible to fit a model of the relationship between these two variables.

1.2 State hypotheses

Remembering back to the lecture and tutorial, the general equations for our hypotheses are:

$$H_0 : \beta_1 = 0$$

$$H_1 : \beta_1 \neq 0$$

In the context of our data, the hypotheses would be:

H_0 : Slope = 0; fertiliser is not a significant predictor of yield.

H_1 : Slope $\neq$ 0; fertiliser is a significant predictor of yield.

If $P > 0.05$, we fail to reject the null hypothesis that the true slope (β_1) is equal to 0. If this is the case, it means our model does not predict better than the mean of our observations, and so there is no advantage to using our model over the mean of y ($\bar{y}$).

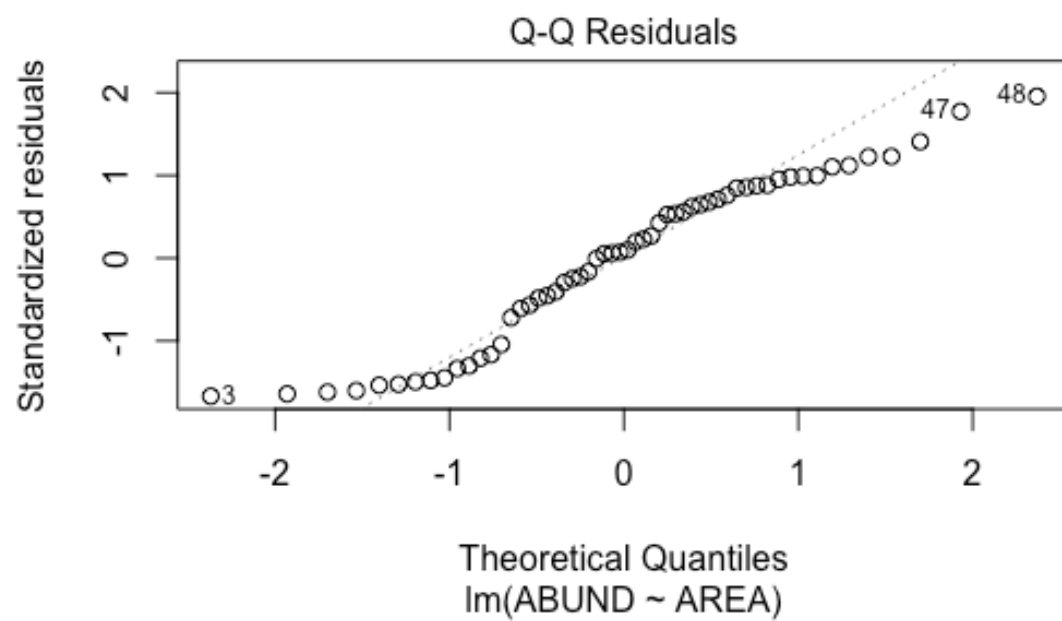
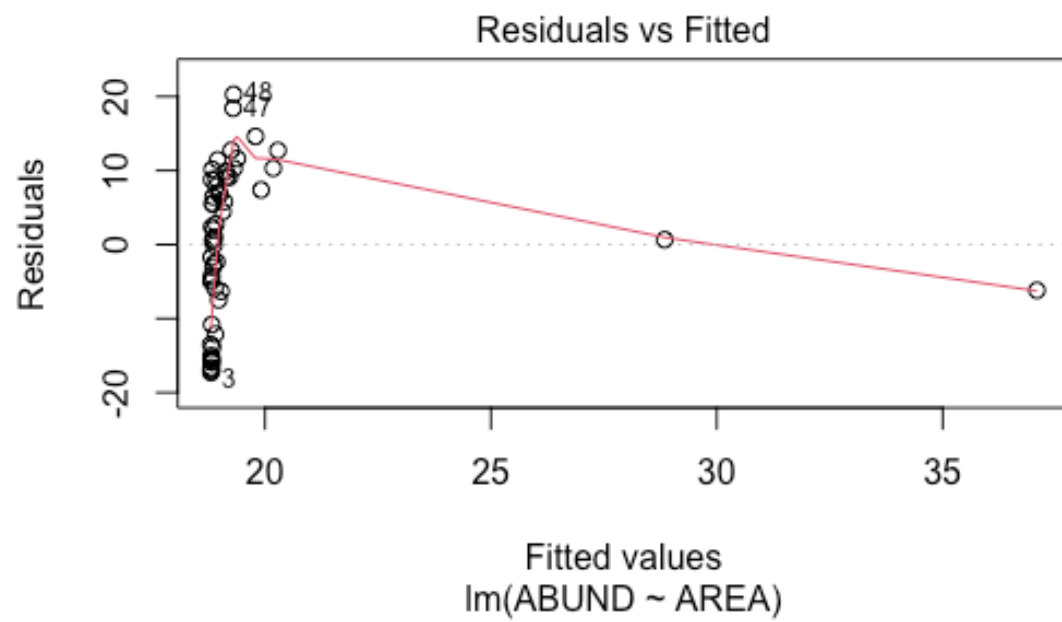
If we find there is a high probability of the slope not being equal to 0 ($P < 0.05$), we can reject the null hypothesis and conclude our model is better at predicting than the mean of our observations.

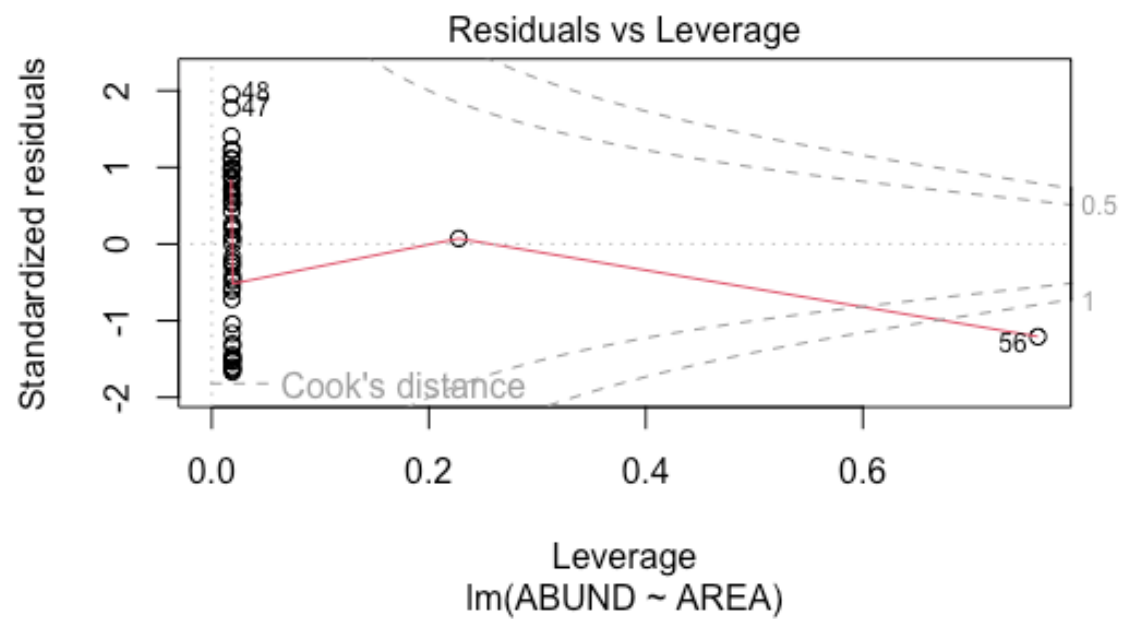
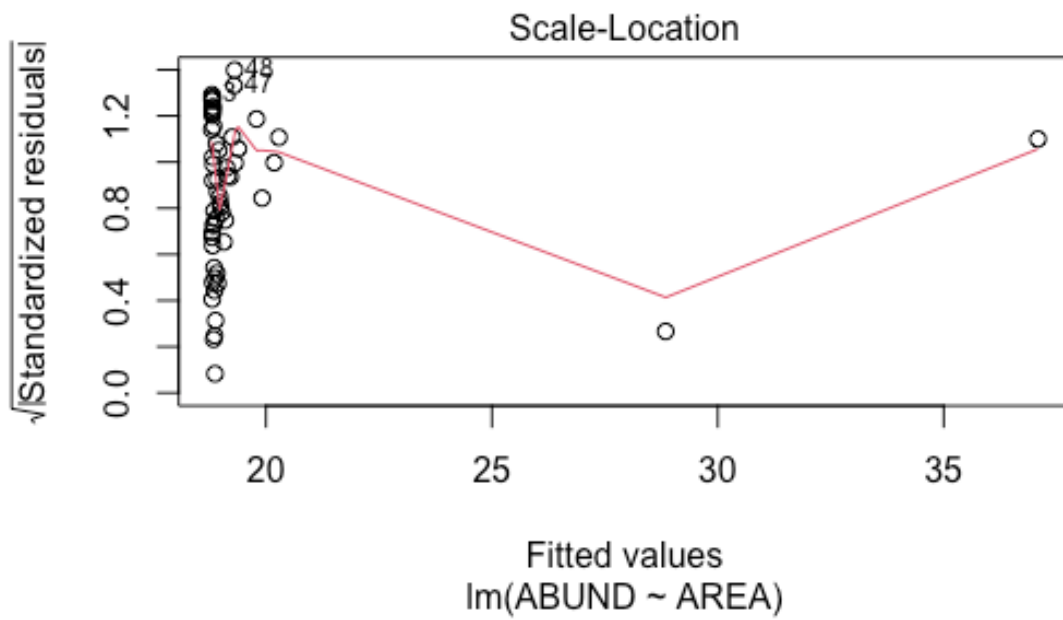
Now we understand what we are testing for, we can fit the model.

1.3 Fit a simple linear regression model

CODE

```
# Fit the model
loyn_lm <- lm(ABUND ~ AREA, data = loyn)
plot(loyn_lm)
```





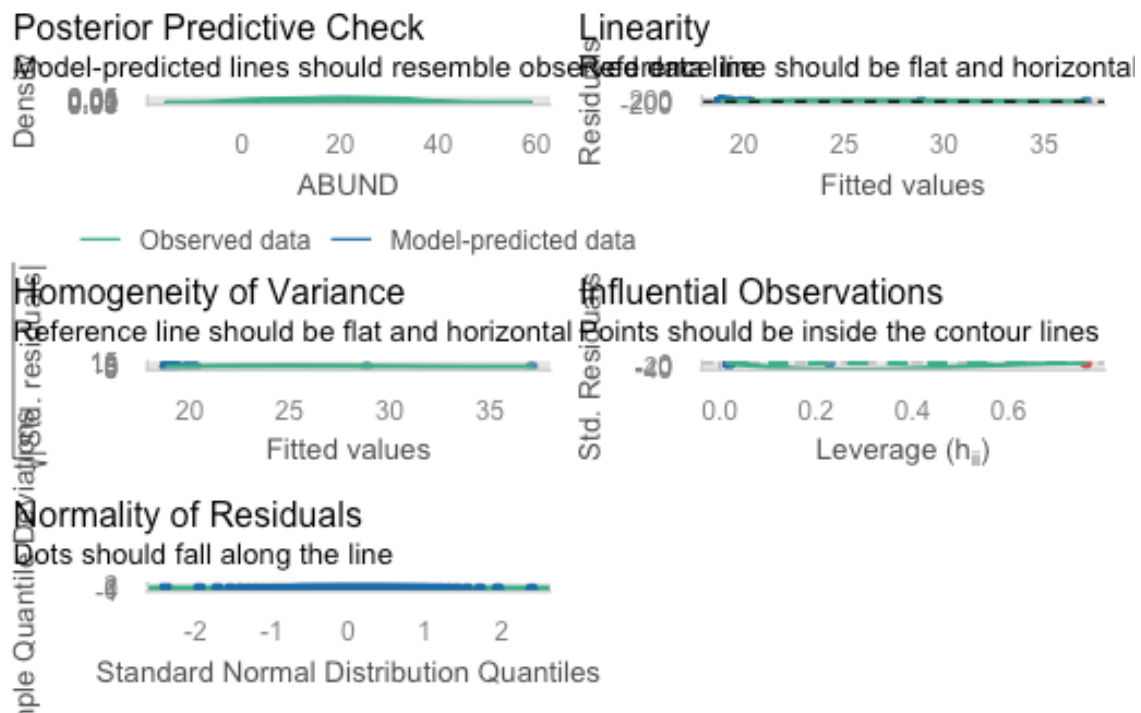
CODE

```
# Look at the model suLog()# Look at the model summary
#summary(loyn_lm) NOT YET
```

The model summary gives us a lot of information about the model fit. We can see that the coefficient for $\log(\text{AREA})$ is 0.26, which means that for every unit increase in $\log(\text{AREA})$, we expect bird abundance to increase by 0.26 on average.

1.4 Check model assumptions

```
CODE
check_model(loyn_lm)
```



The diagnostic plots show us that the residuals are not normally distributed, and that there is some heteroscedasticity (the variance of the residuals is not constant). This suggests that the model may not be a good fit for the data, and that we may need to consider a different model or transformation of the data.

1.5 Interpret model parameters and fit statistics

1.6 Make conclusions

Exercise 2: California streamflow

The following dataset contains 43 years of annual precipitation measurements (in mm) taken at (originally) 6 sites in the Owens Valley in California. I have reduced this to three variables labelled lake_sabrina (Lake Sabrina), pine_creek (Big Pine Creek), rock_creek (Rock Creek), and the dependent variable stream runoff volume (measured in ML/year) at a site near Bishop, California (labelled runoff_volume).

Note the variables have already been log-transformed to increase normality of the residuals in the regressions.

Start with a full model and manually remove the variables one at a time, checking every time whether removal of a variable actually improves the model.

CODE

```
# read in the data
stream_data <- read_xlsx("data/california_streamflow.xlsx", "streamflow")
names(stream_data)
```

OUTPUT

```
[1] "lake_sabrina" "pine_creek"    "rock_creek"    "runoff_volume"
```

CODE

```
s.mod_full <- lm(runoff_volume ~ lake_sabrina + pine_creek + rock_creek, data=stream_data)
s.mod_full <- lm(runoff_volume ~ ., data=stream_data) ## you can also use the . to indicate use all
variables
summary(s.mod_full)
```

OUTPUT

```
Call:
lm(formula = runoff_volume ~ ., data = stream_data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.09885 -0.03331  0.01025  0.03359  0.09495

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.25716    0.12360   26.352 < 2e-16 ***
lake_sabrina   0.05631    0.03756    1.499  0.14185
pine_creek     0.21085    0.06756    3.121  0.00339 **
rock_creek     0.43838    0.08798    4.983 1.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.04861 on 39 degrees of freedom
Multiple R-squared:  0.8817,    Adjusted R-squared:  0.8726
F-statistic: 96.88 on 3 and 39 DF,  p-value: < 2.2e-16
```

Partial F-Tests

The above analysis tells us that both `pine_creek` & `rock_creek` are significant, according to the t-test, in the model and `lake_sabrina` is not? This involves performing Partial F-Tests as discussed in the lecture.

This can be done in **R** by using `anova()` on two model objects. To be able to compare the models and run the anova, you need to make objects of all the possible model combinations you want to compare.

CODE

```
s.mod_reduced <- lm(runoff_volume ~ rock_creek + pine_creek, data=stream_data)
anova(s.mod_reduced, s.mod_full)
```

The last row gives the results of the partial F-test.

Question 1

Should we remove `lake_sabrina` from the model?

Question 2

Is the p-value for the f-test the same as for the t-test?

Question 3

Write out the hypotheses you are testing.

Perform a Partial F-Test to work out if the removal of `lake_sabrina` and `pine_creek` improves upon the full model.

CODE

```
s.mod_reduced2 <- lm(runoff_volume ~ lake_sabrina + pine_creek, data=stream_data)
anova(s.mod_reduced2, s.mod_full)
```

OUTPUT

Analysis of Variance Table

```
Model 1: runoff_volume ~ lake_sabrina + pine_creek
Model 2: runoff_volume ~ lake_sabrina + pine_creek + rock_creek
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     40 0.150845
2     39 0.092166  1    0.05868 24.83 1.321e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4

Which variable should be added to the model containing `rock_creek`?

Question 5

Could things be even simpler? Perform a partial F-Test to see if a model containing `rock_creek` alone could be suitable.

CODE

```
s.mod_reduced3 <- lm(runoff_volume ~ rock_creek, data=stream_data)
anova(s.mod_reduced3, s.mod_full)
```

Question 6

What is your optimal model?

Review

- Simple linear regressions model the relationship between two variables
 - We can also make linear models with more than one predictor
- We can use histograms and correlation matrices to do some preliminary exploration of the data
- If any variables are skewed, we can transform them
- Looking at a correlation matrix to identify the best predictors (for both simple and multiple linear regression)
- Fit model using `lm()` function
- Check assumptions:
 - Collinearity (multiple linear regression only)
 - Linearity
 - Independence
 - Normality
 - Equal variance
- Use `summary()` to look at model output and interpret it
 - F-test : overall model significance
 - Coefficients table : individual predictors' significance
 - R^2 : How much variation in the data is explained by the model?

That's it for today! Great work fitting simple and multiple linear regression! Next week we jump into stepwise selection!

Attribution